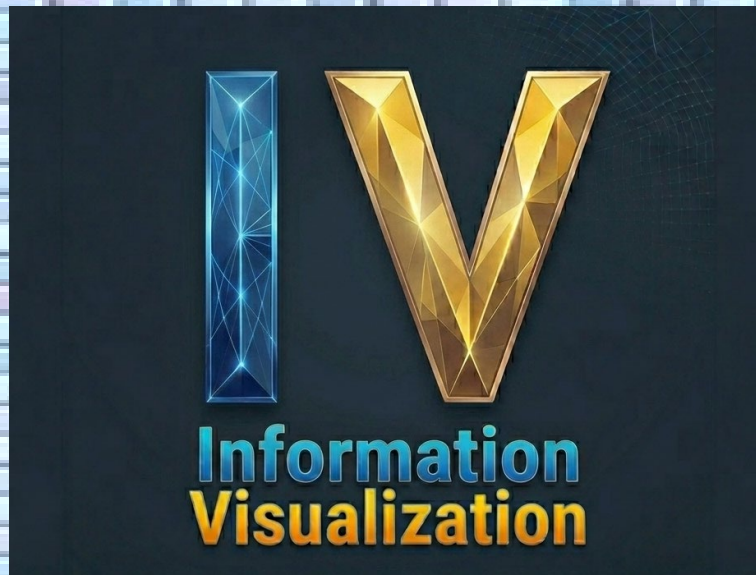


Vorlesung
Informationsvisualisierung

Kapitel 3 VISUELLE WAHRNEHMUNG



Lecture
Information Visualization

Chapter 3 VISUAL PERCEPTION

TODAY:
Lots of
slides



Nihil est in intellectu quod non erat in sensu*

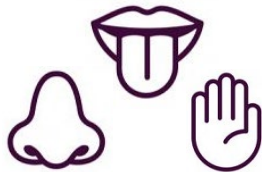
3



75% of real world information is **visually** perceived.



13% acoustically



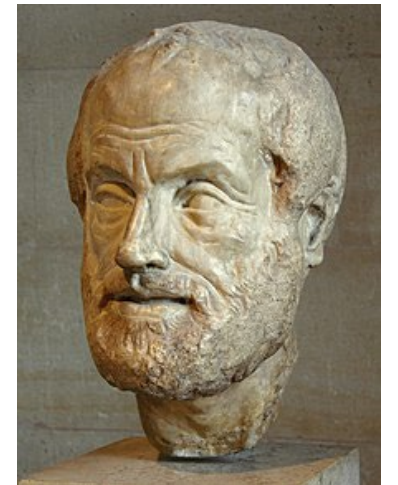
remaining **12%**: other senses

*

There is nothing in the understanding that was not earlier in the senses.

(Nichts ist im Verstand, das nicht zuvor in den Sinnen war.)

Aristoteles 384-322 v. Chr



Chapter
3.1

Visual Memory

Visual Memory

5



- High recognition rate of long term memory
- Shepard, R. N. (1967):
 - ▣ Test persons looked through a series of about 600 images, words or sentences. Then they were tested for their ability to recognize these.
 - ▣ Error rates: 1,5% for images, 10% for words and 11,8% for sentences.
- Standing, L. (1973)
 - ▣ Showed 10.000 images to test persons.
 - ▣ Error rate: 17%.

Change Blindness

6



Change Blindness

7



Visual Memory

8

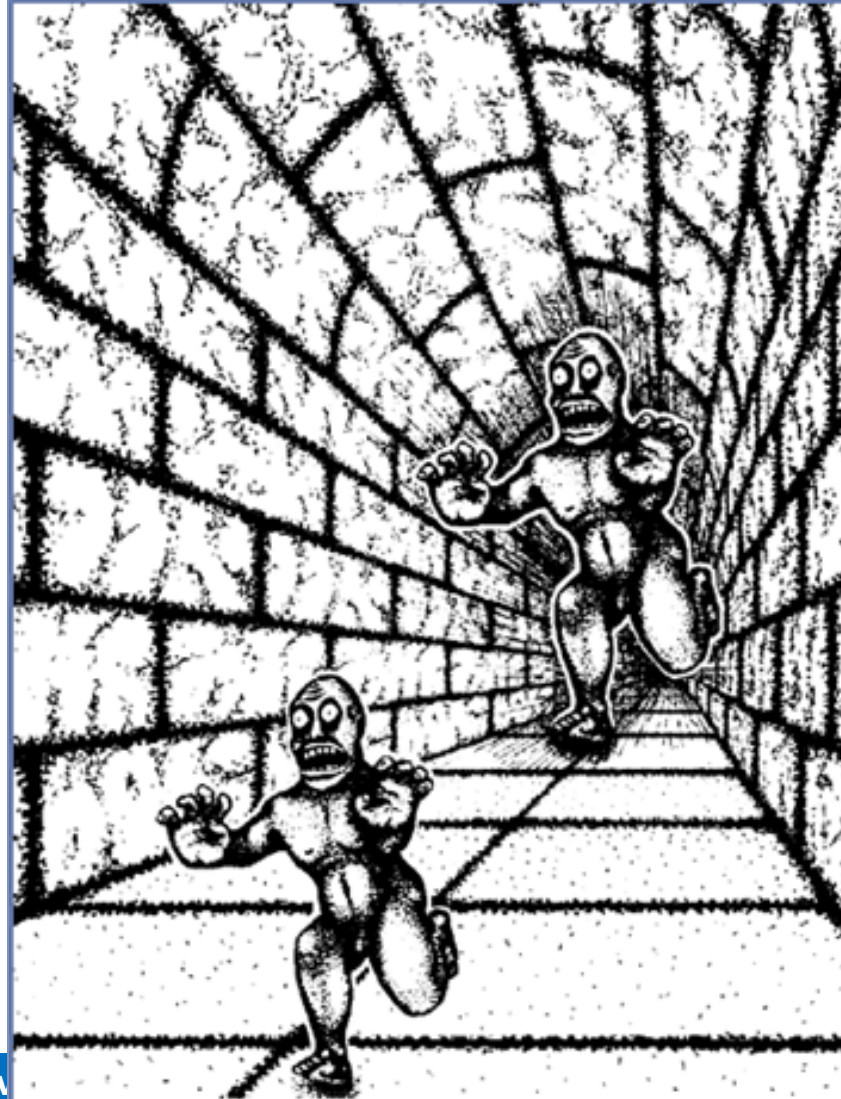
- **Sensory Memory:** Fractions of a second to a few seconds
- **Short-Term Memory (STM):** 15 to 30 seconds
- **Long-Term Memory (LTM):** Anything beyond 30 seconds (minutes, days, years, or a lifetime)

Chapter
3.2

Visual Confusion

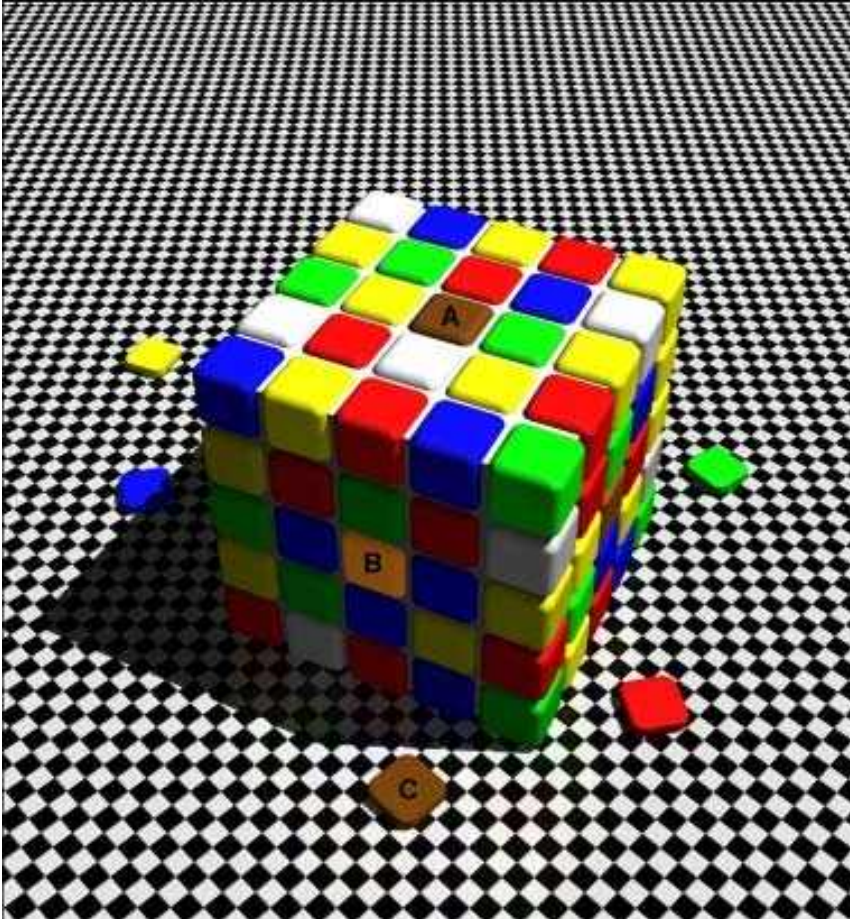
Who is the bigger monster?

10



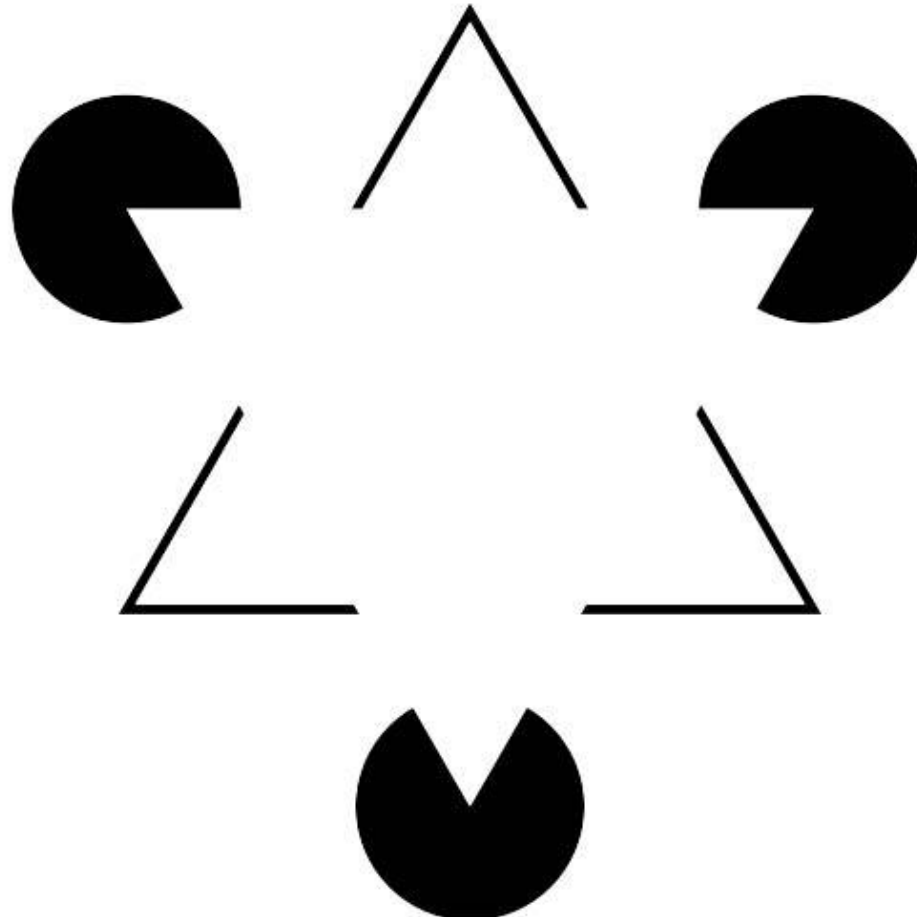
What color is brighter, A or B?

11



How many triangles?

12



Spot Spotty!

13



One glass too many

14



Content warning: *This image does not show sexual content, but maybe misperceived as such.*

The human brain is highly sensitive to recognizing biological forms and patterns.



Observations

15

- Conversation partner can be swapped unnoticed
- Size and color are wrongly assessed
- Contours are automatically completed

Visual perception is selective and interpreting.

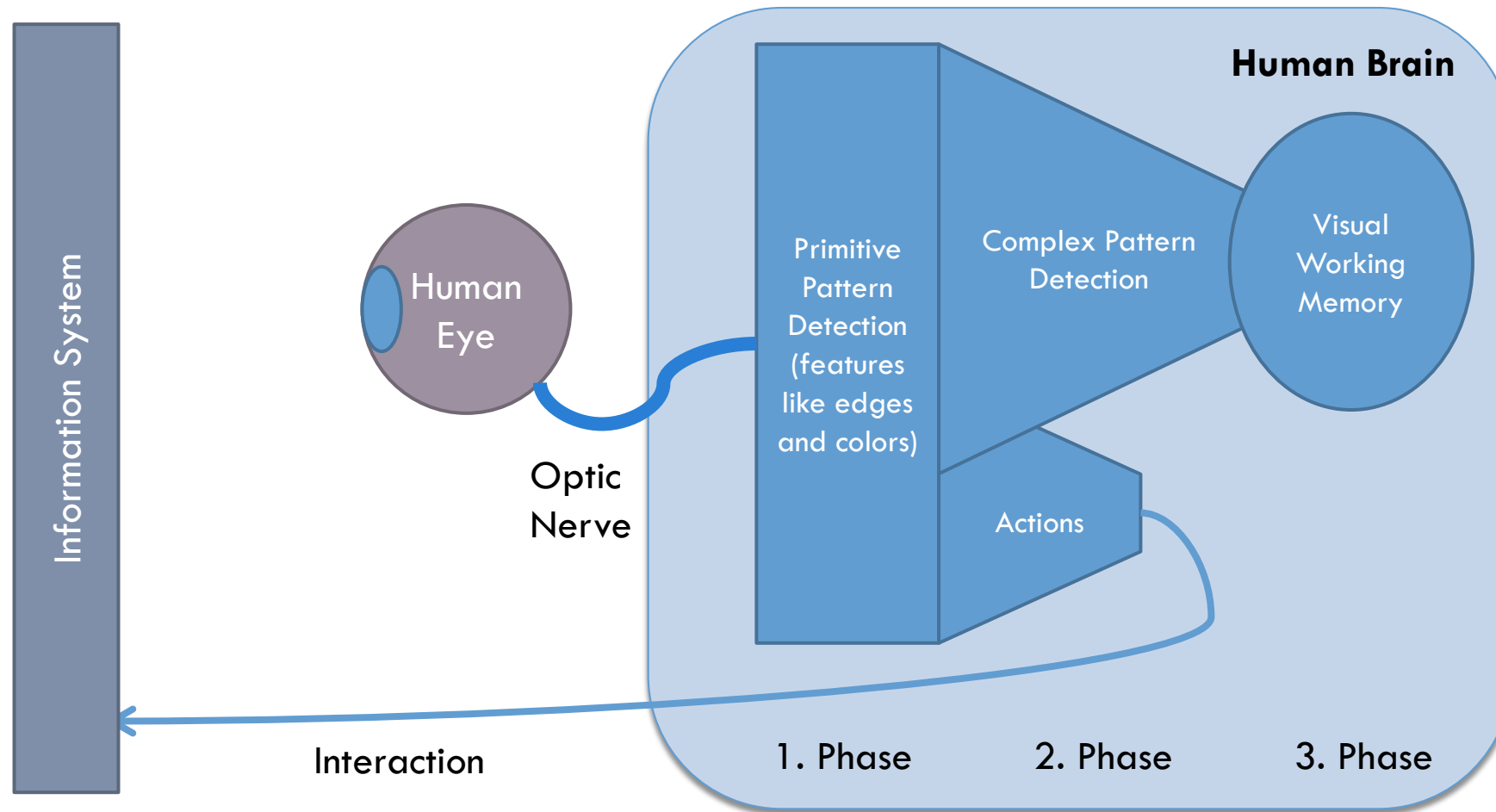
... and sometimes misleading!

Chapter
3.3

Visual Information Processing

Visual Information Processing

17



Three-stage model of human visual information processing by Colin Ware

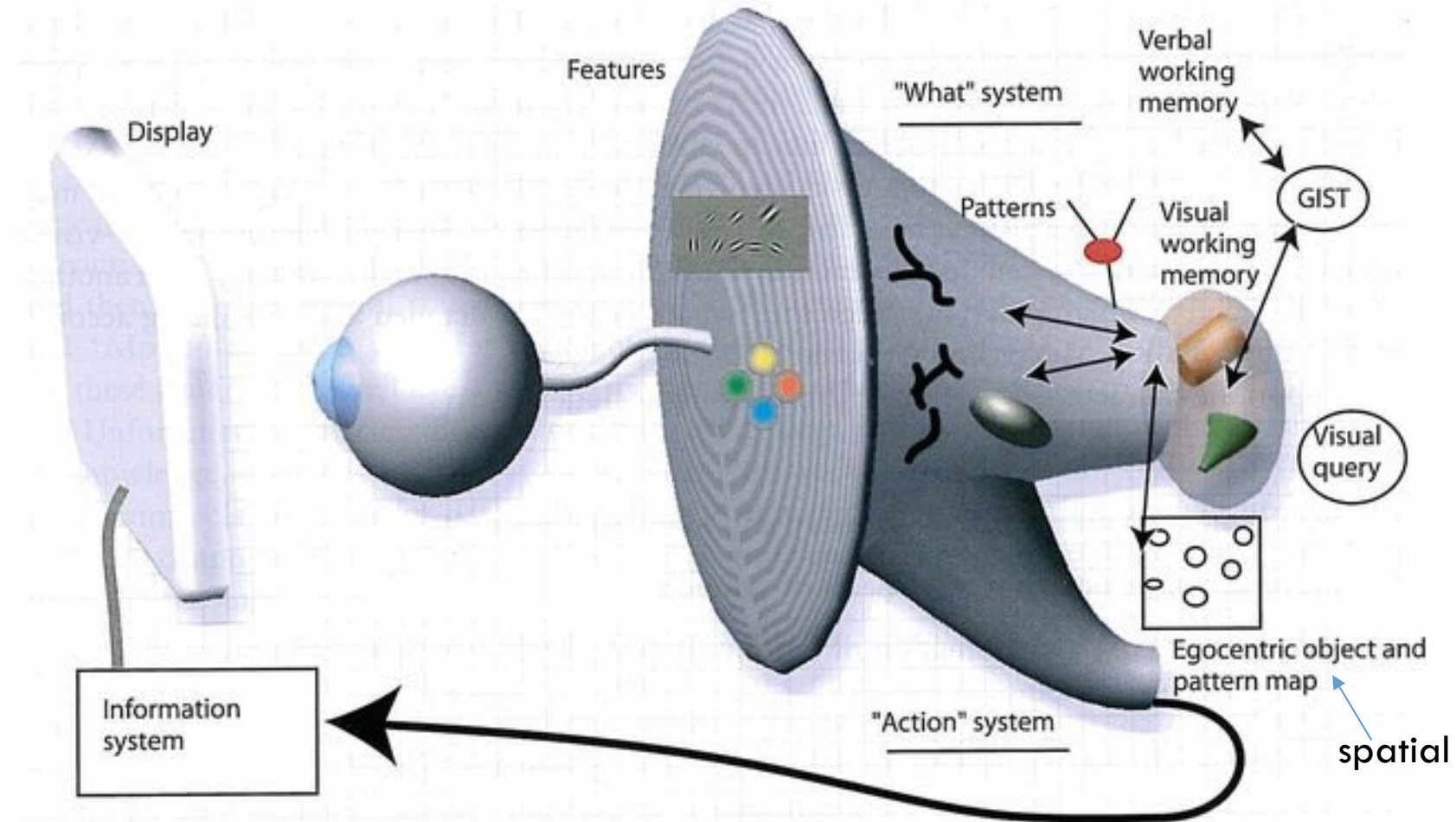
Visual Information Processing

18

1. Phase

2. Phase

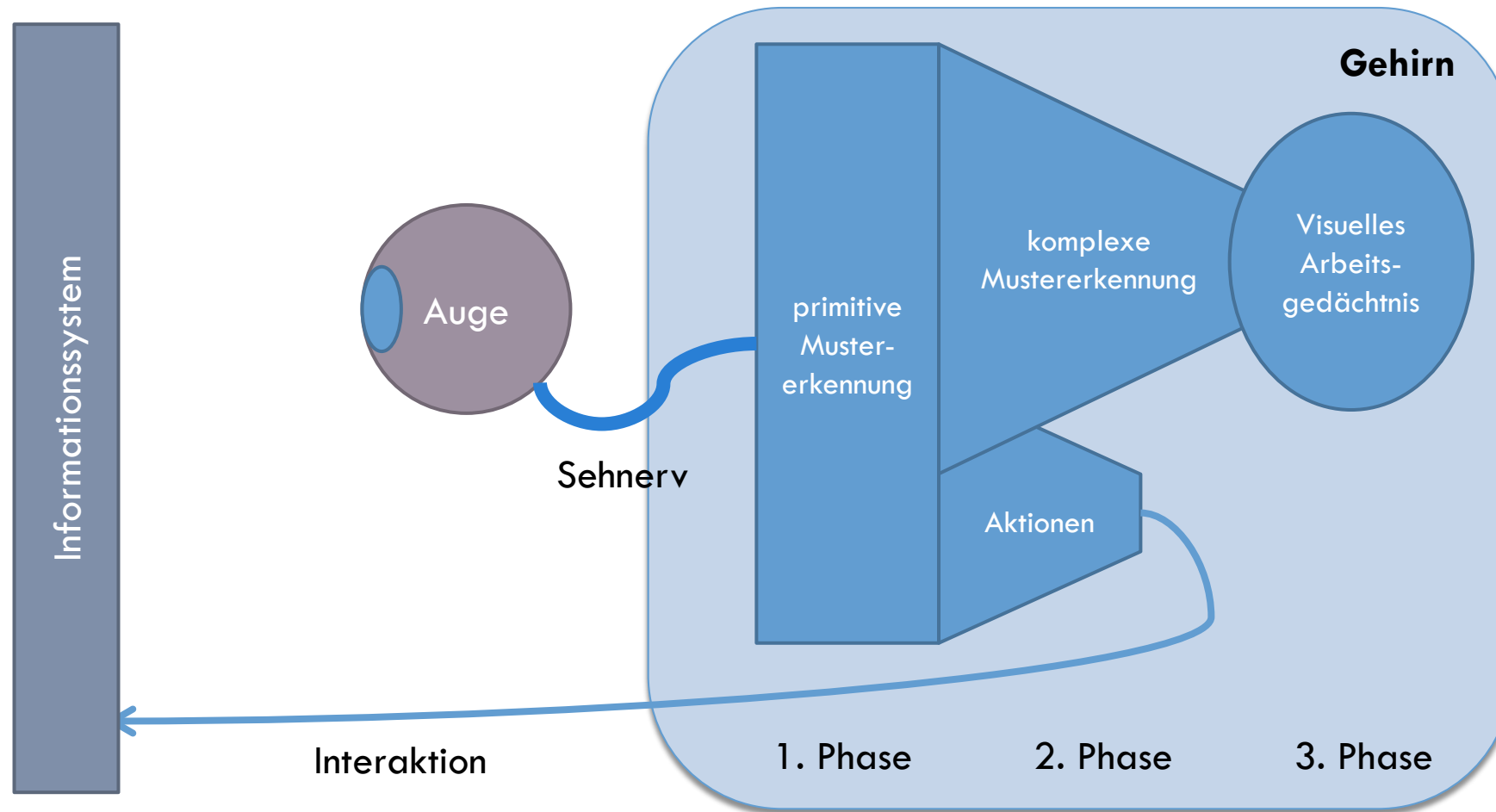
3. Phase



Three-stage model of human visual information processing by Colin Ware

Visuelle Informationsverarbeitung

19



Dreiphasen-Modell nach Colin Ware

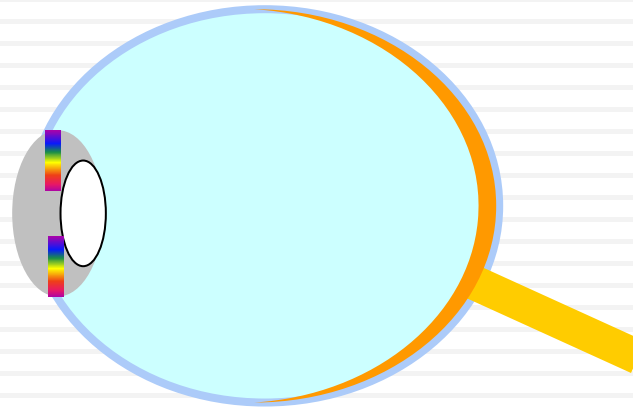
Literature

20

- Colin Ware: *Information Visualization, Second Edition: Perception for Design (Interactive Technologies)*, Morgan Kaufmann 2004
- Colin Ware: *Visual Thinking for Design*, Elsevier 2008

Chapter
3.4

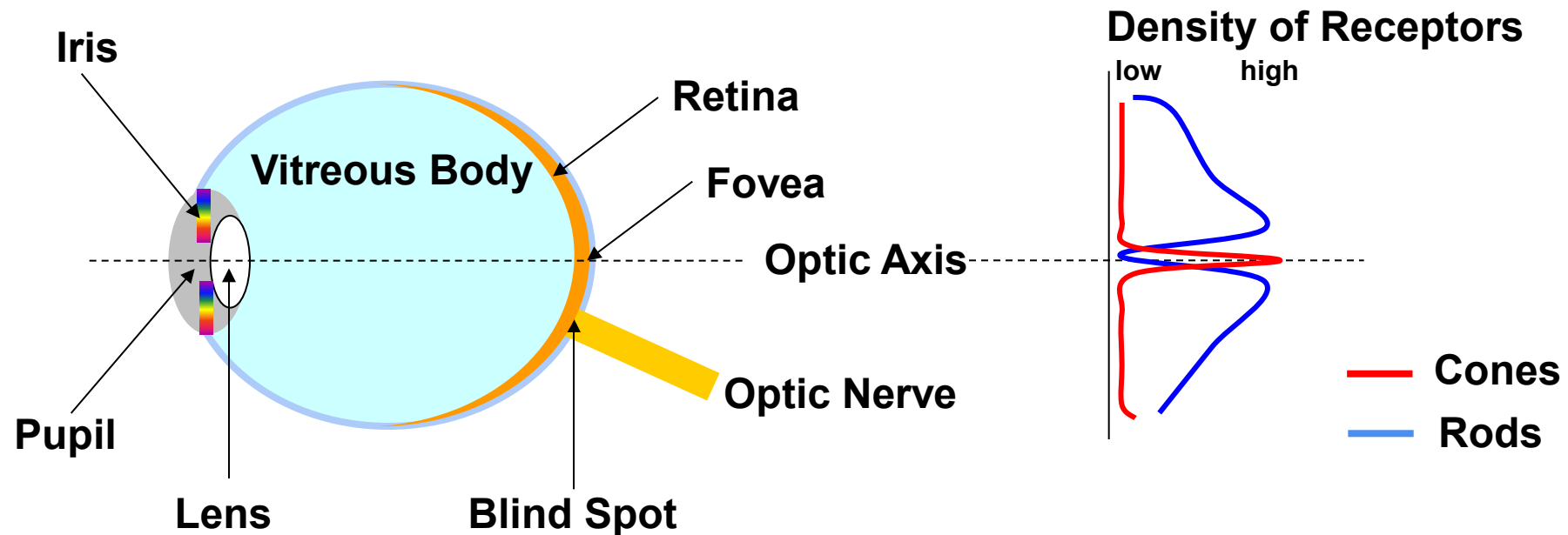
Human Eye

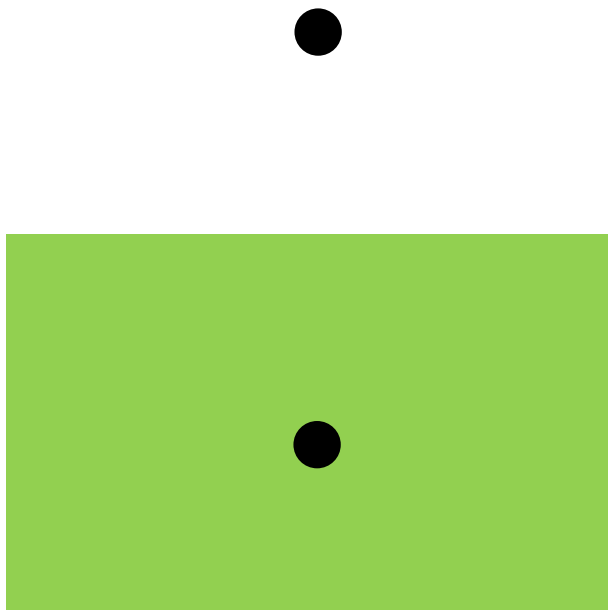


The Human Eye

22

- In the center of the retina is the point of best vision (Fovea centralis).
- In the retina there are two kinds of receptors:
 - ▣ cones (color vision)
 - ▣ and rods (b/w vision)



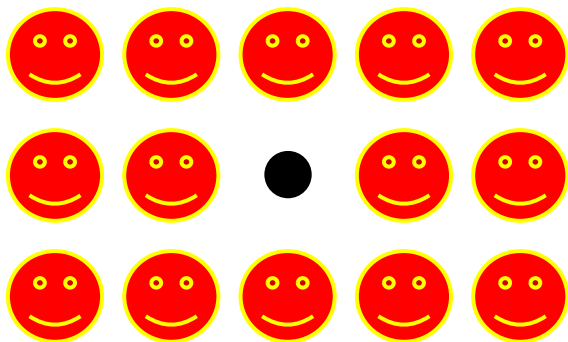


Cover right eye



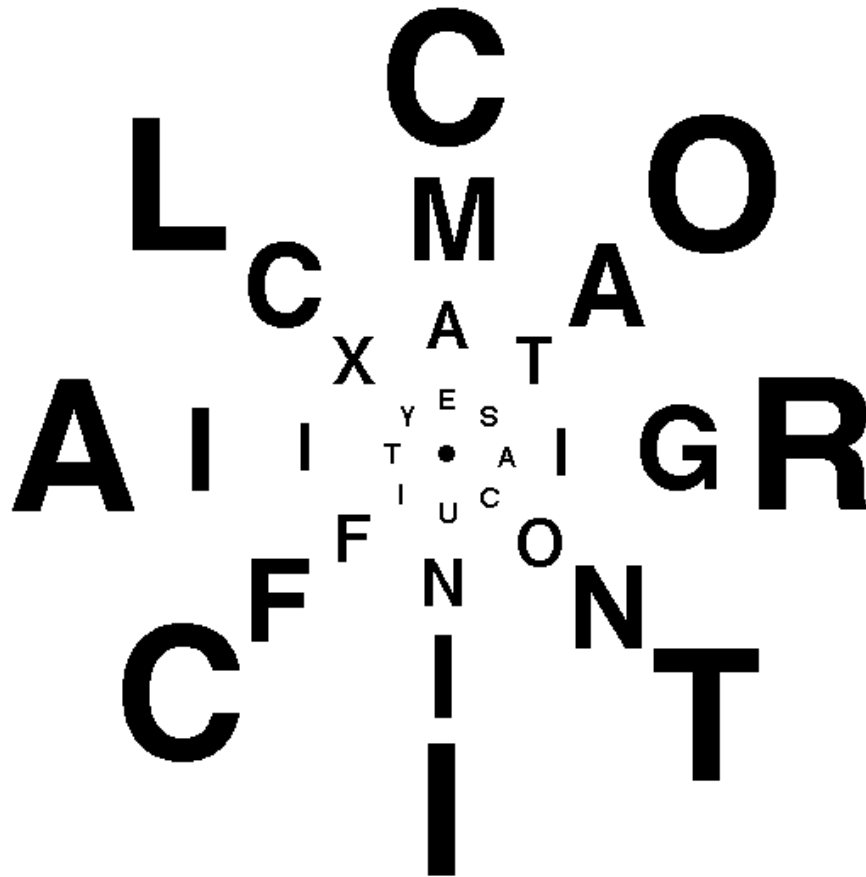
Focus + with left eye

Move closer or further
away from the slide



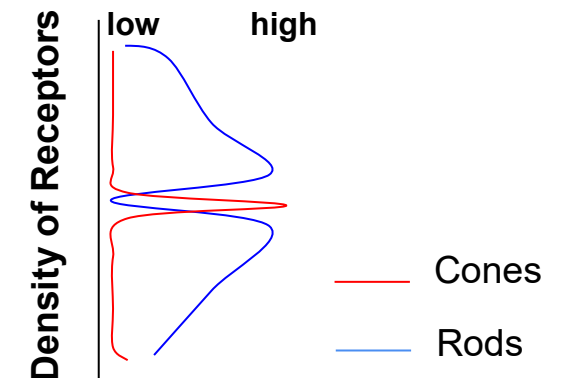
Peripheral Acuity

24



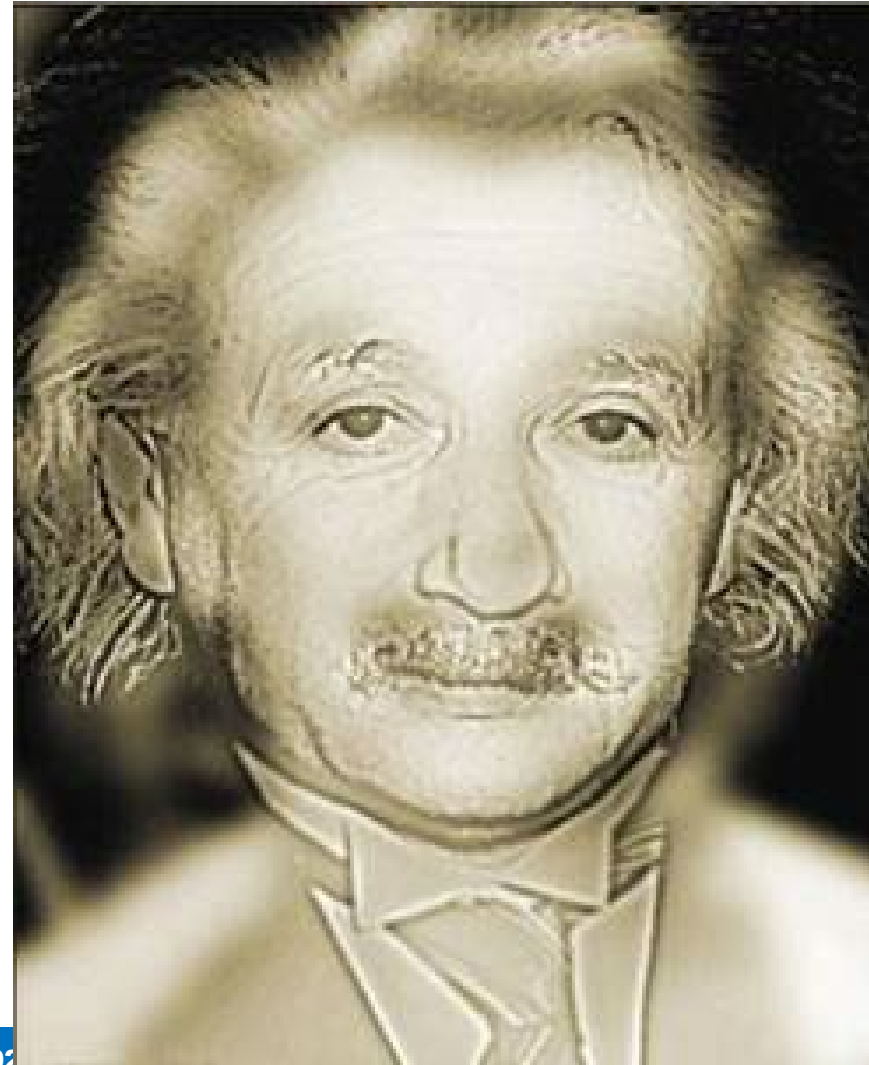
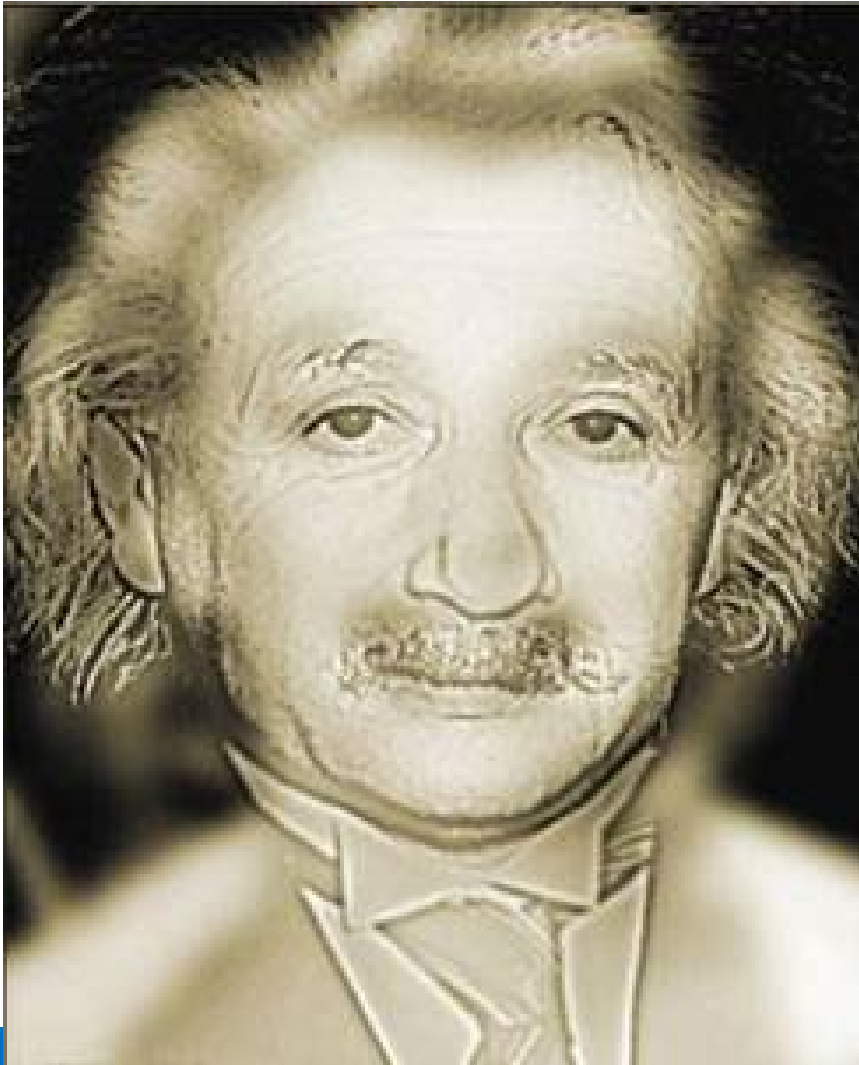
With strict fixation of the center spot, each letter is equally legible (Anstis, 1974)

- Center
 - ▣ sharp
 - ▣ color and brightness
- Periphery
 - ▣ unsharp (blurred)
 - ▣ only brightness



Resolution matters

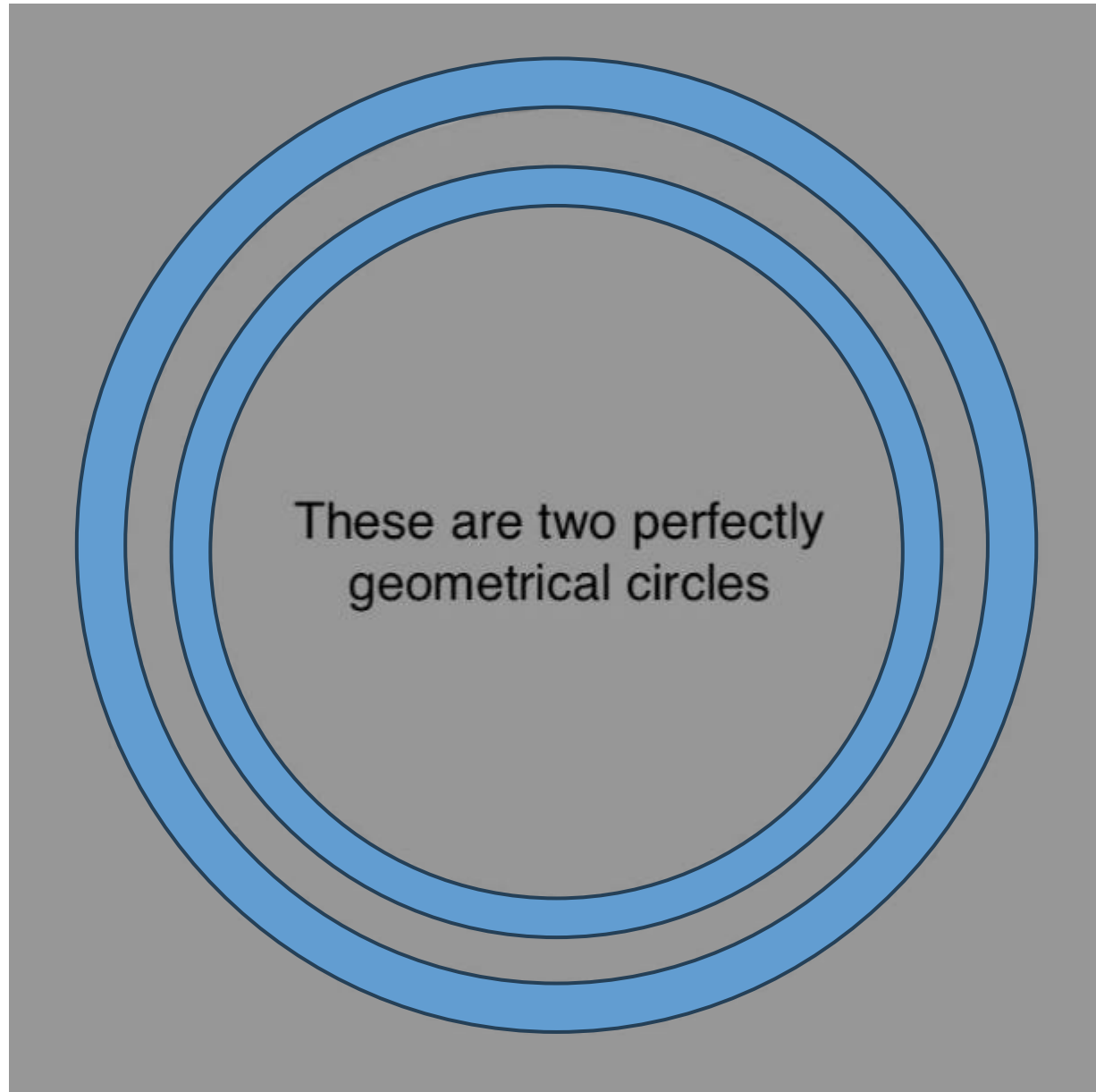
25



Eye movements

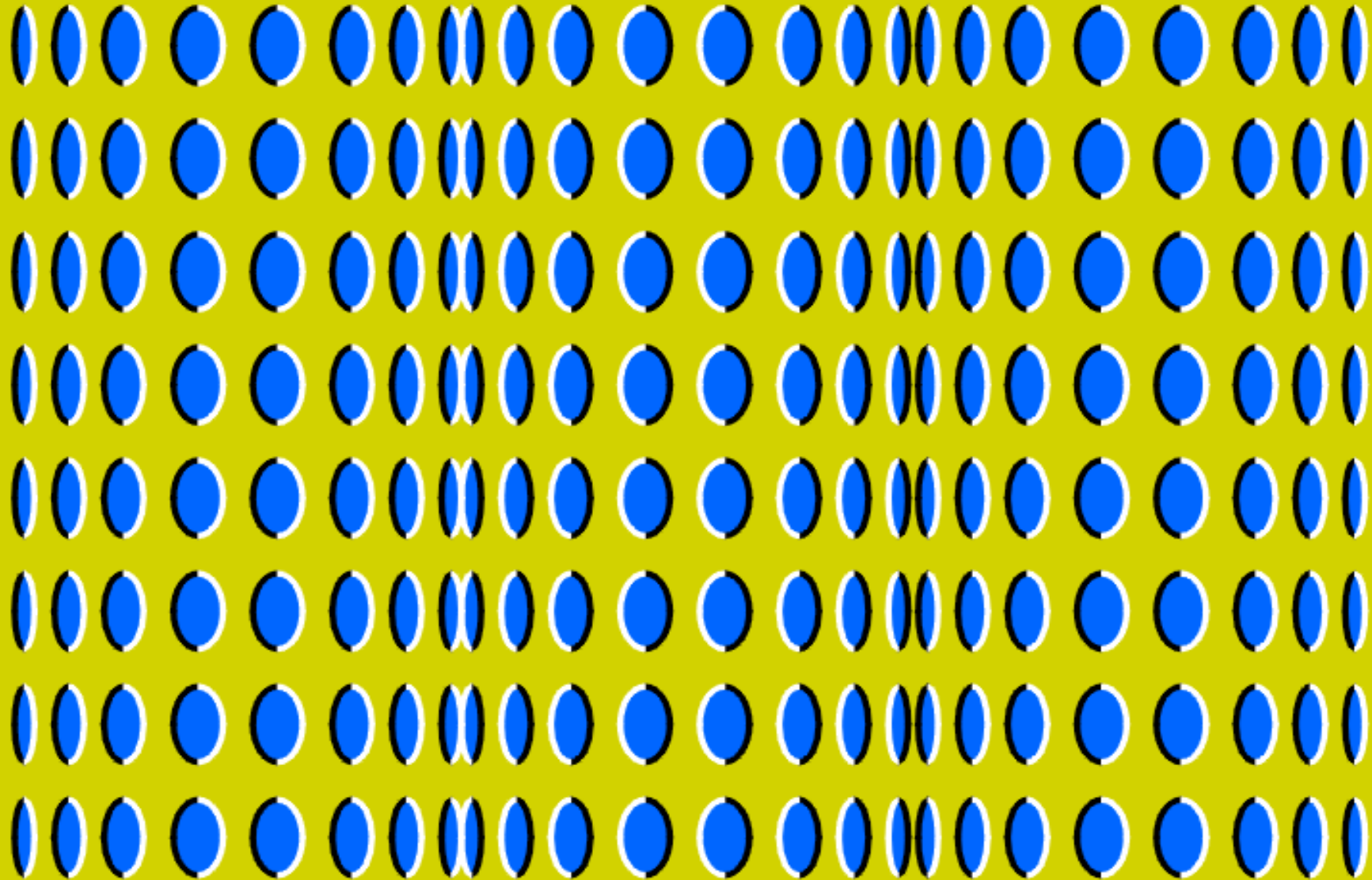
26

- Fast eye movements (Saccade)
 - ▣ Initialization: 200ms
 - ▣ Duration ca. 50ms; Speed up to $900^{\circ}/s$
 - ▣ No visual perception during movement !!!
- Fixation
 - ▣ Exact observation / no eye movement
 - ▣ Visual perception takes place
- Eye movements during an activity
 - ▣ Goal-oriented: only relevant objects are focused
(→ Change Blindness)
 - ▣ Predictive: what will happen next, what will be of interest next

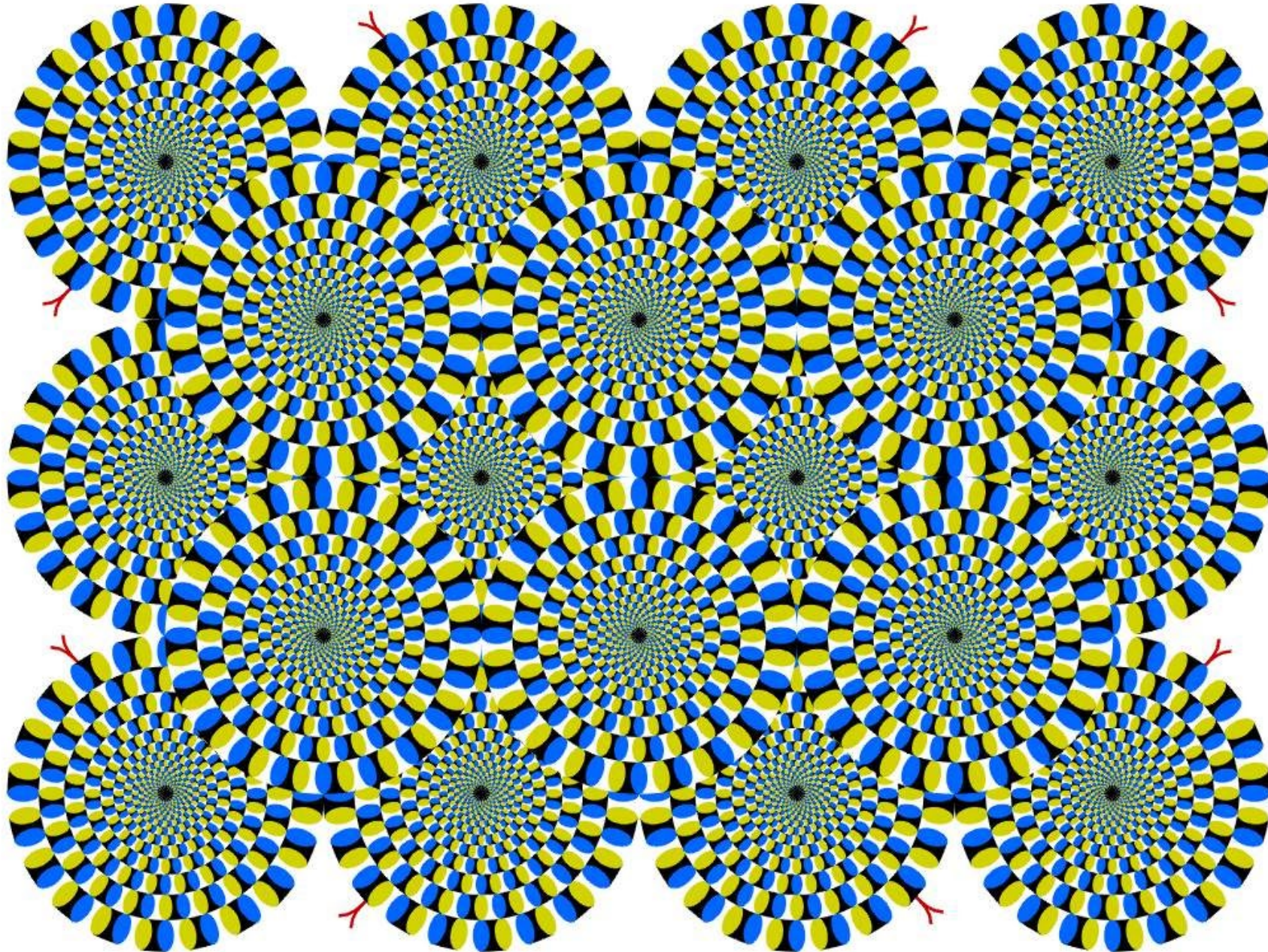


But it does move. (Und sie dreht sich doch) Galileo Galilei

28



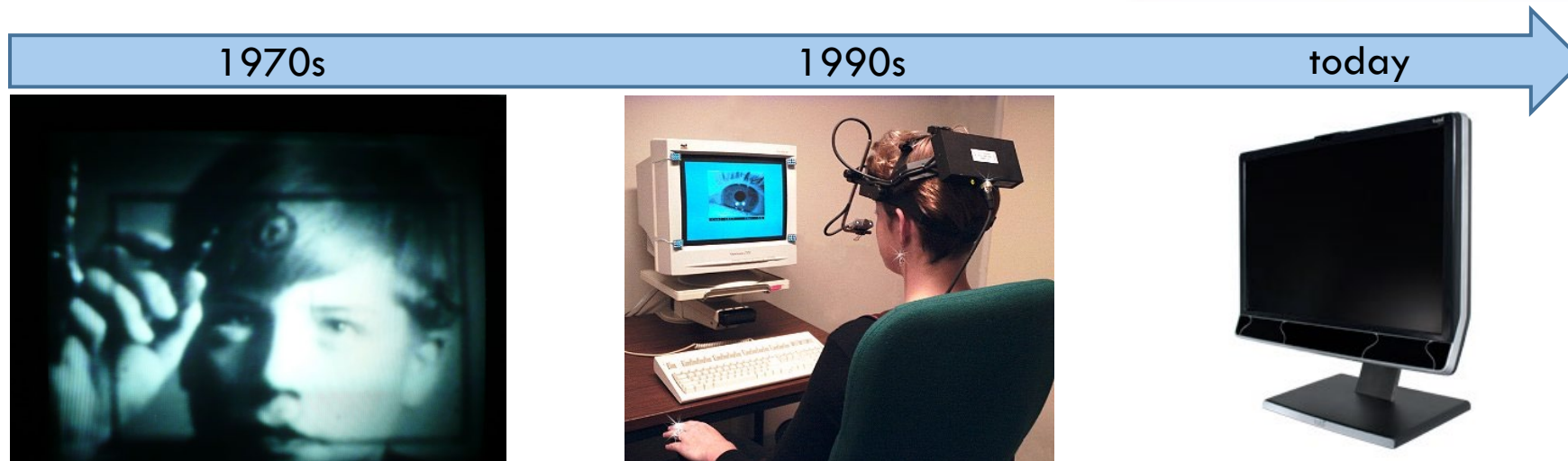
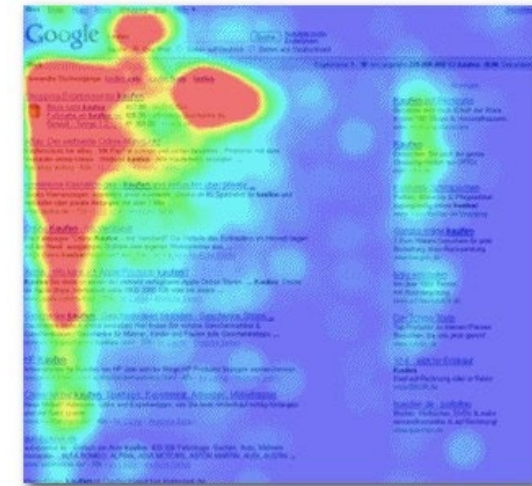
But it does move. (Und sie dreht sich doch) Galileo Galilei



Eye Tracking

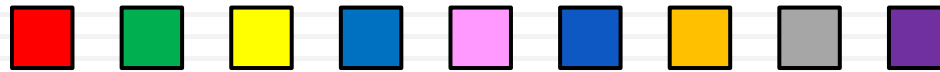
30

- Capturing of eye movements
- Applications:
 - ▣ Research
 - ▣ Usability of web sites or software
 - ▣ Input device to control computers

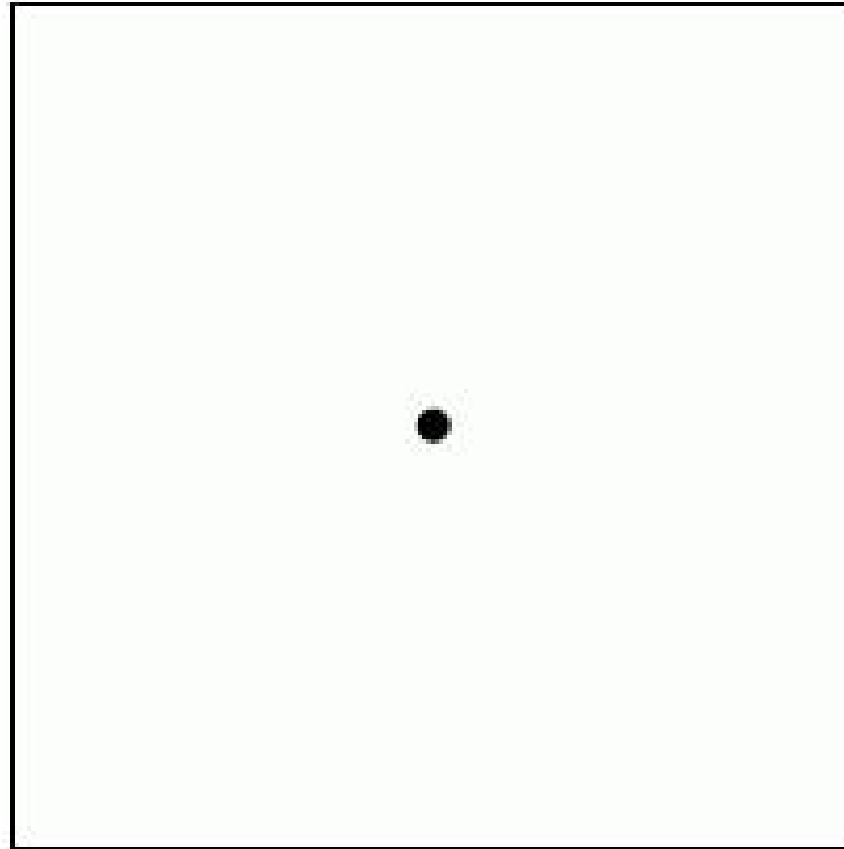
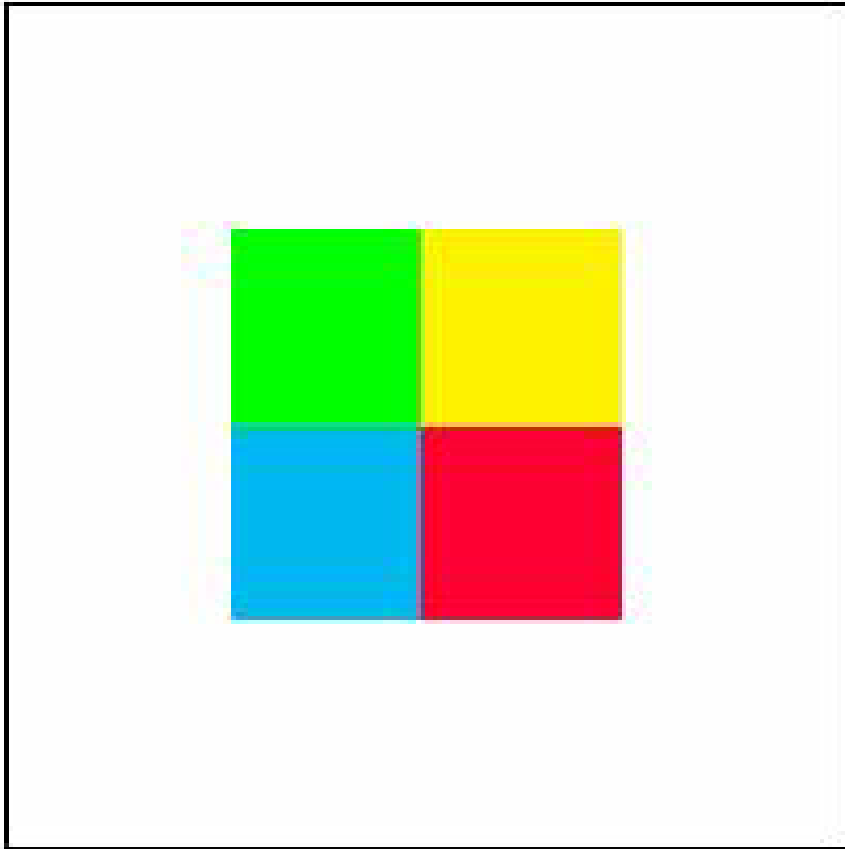


Chapter
3.5

Color Perception



- Look for about 30 seconds at the center of the colored square. Then focus the point in the center of the right square.







Opponent Color Theory [Hering, 1874/1878]

(Gegenfarbtheorie)

Source:

https://en.wikipedia.org/wiki/Opponent_process

35

- Three separate chemical processes in the retina with two *opponent colors* each. It is more efficient for the visual system **to record *differences* between the responses of cones**, rather than each type of cone's individual response.

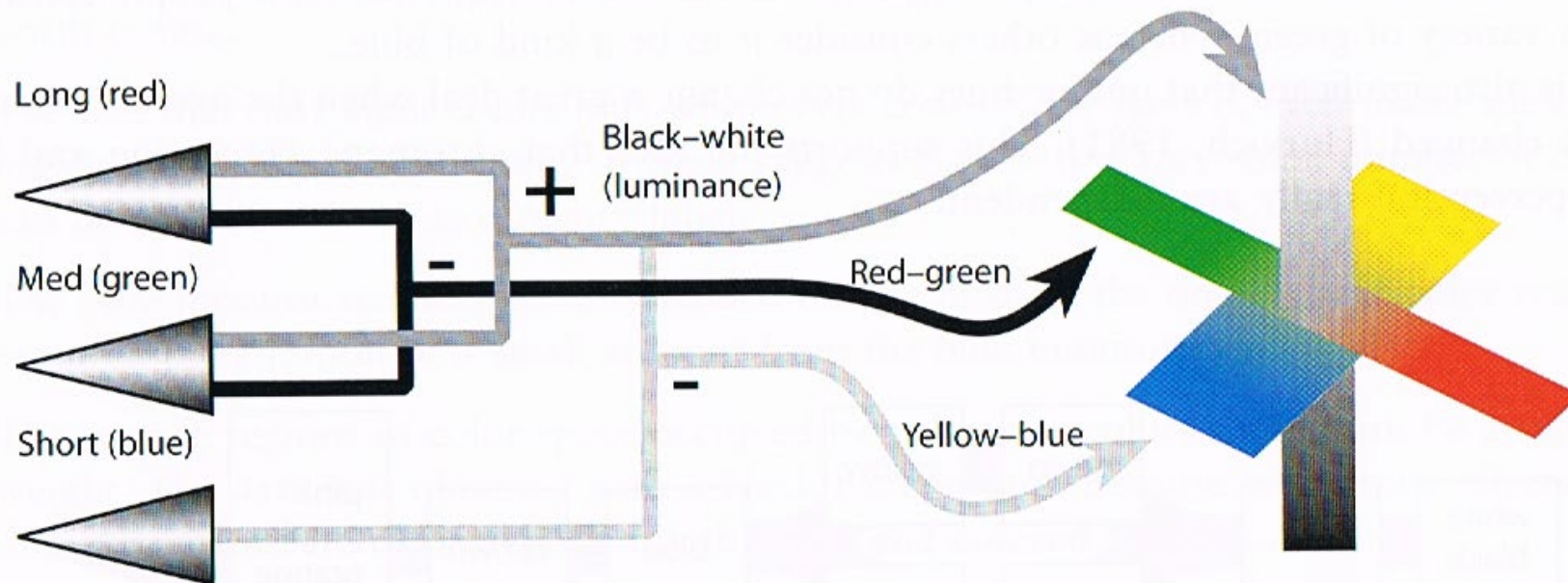
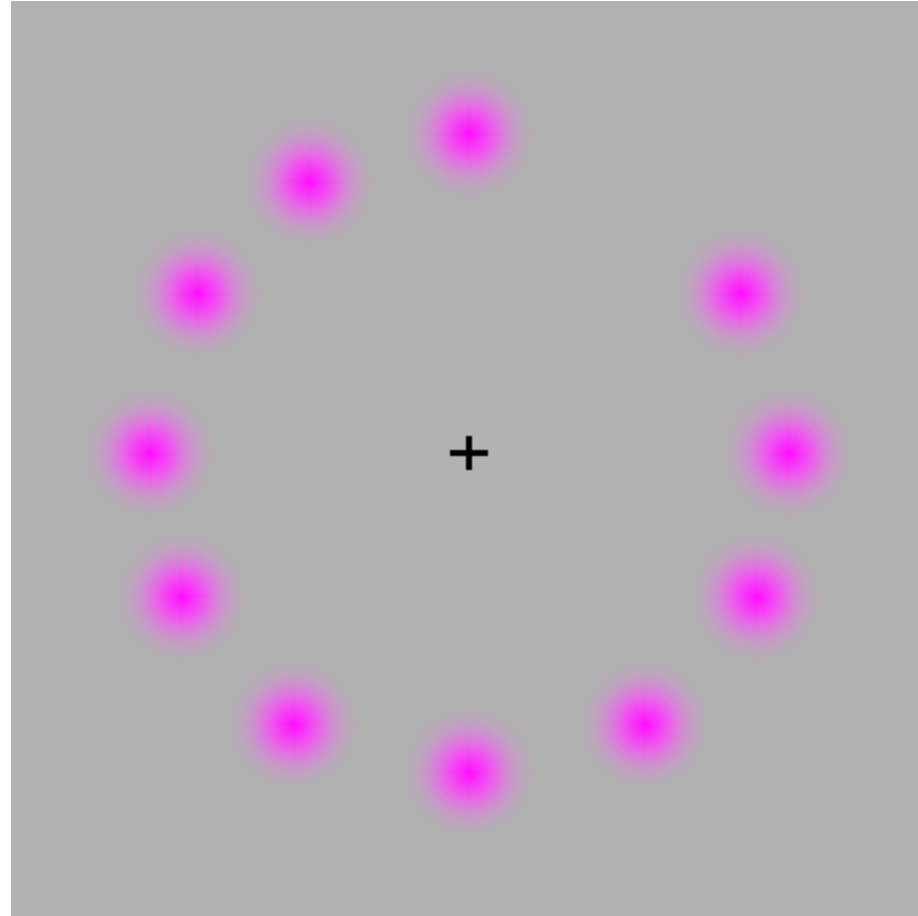


Figure 4.10

In the color opponent process model, cone signals are transformed into black-white (luminance), red-green, and yellow-blue channels.

Another After Image



- The receptors that are constantly exposed to the same stimulus will eventually exhaust resulting in a decrease in signal to the brain.

Color blindness

37



GREEN

Color blindness

38

- Red-green
 - ▣ Germany: males: 9%, females 1%
- Yellow-blue (very rare)

- Total color blindness
 - ▣ No color perception (very rare)

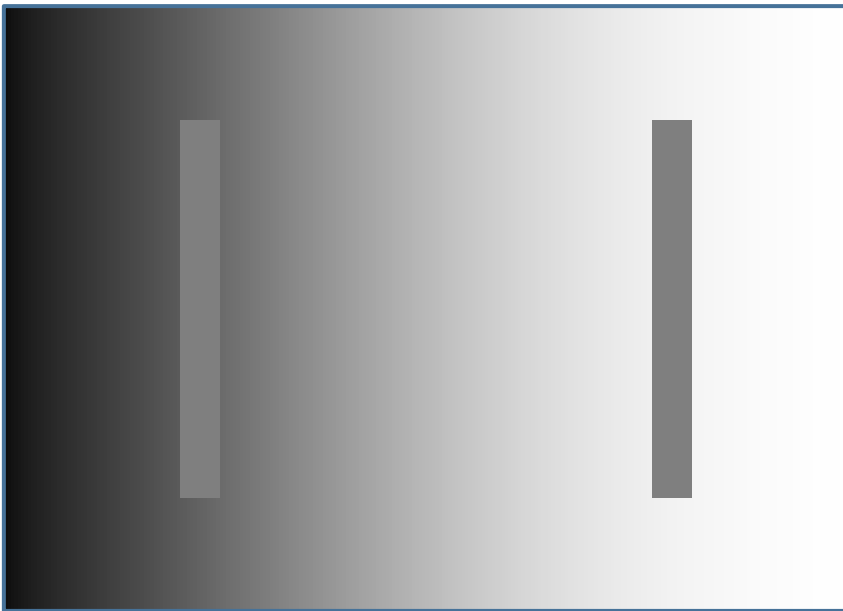


Test for red-green blindness

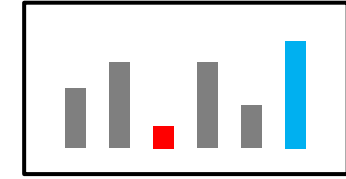
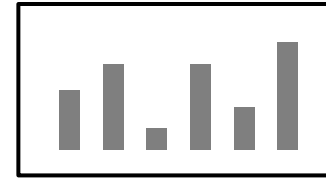
Brightness and color contrast

39

- Differences in brightness can be better perceived than those in color.
- Color and brightness are relative.

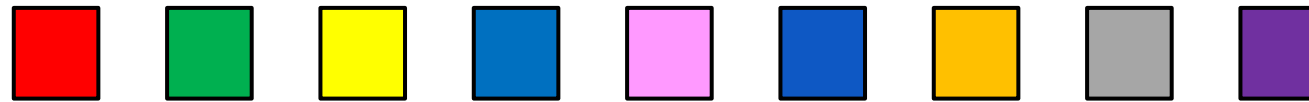


Design Recommendations



40

- Encode important resp. small value differences by brightness.
- Use color to emphasize elements.
- Color coding of categories: maximum 6 to 12 different colors:



- Color scales should simultaneously vary both color and brightness
- Color blindness → Avoid red-green color scales.

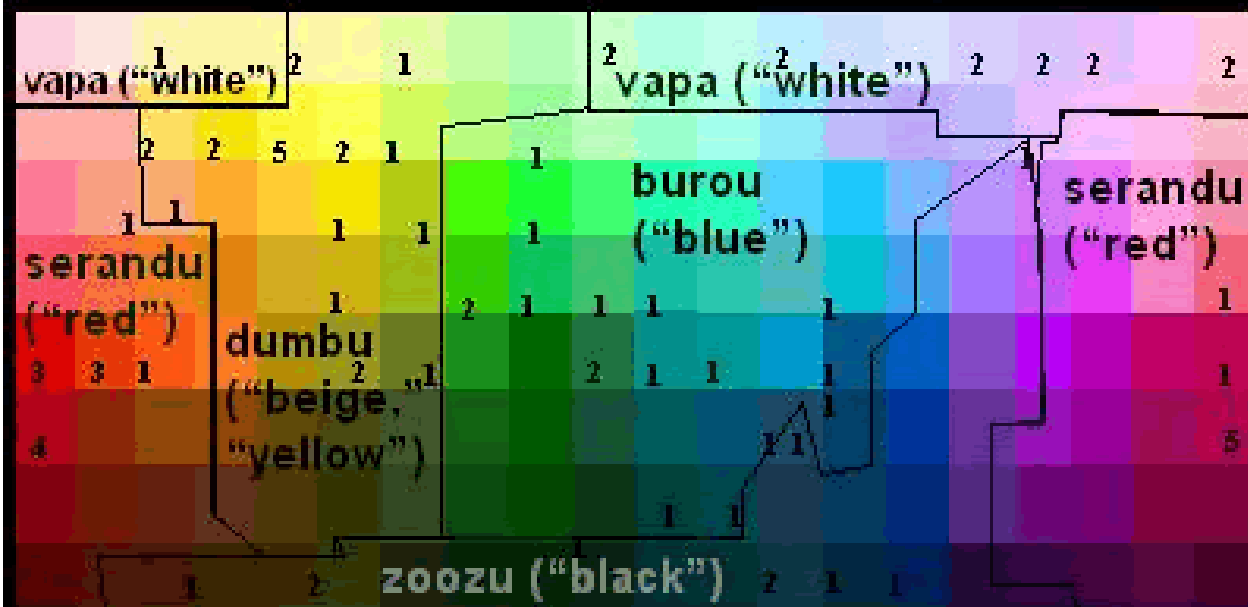


Color perception and human language

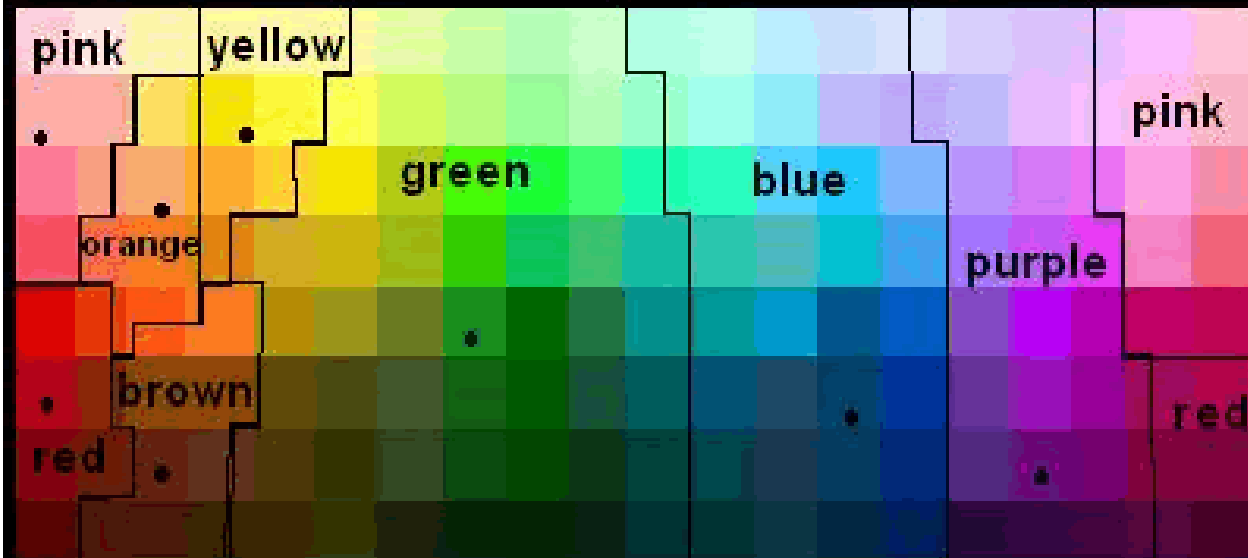
41

- Himba: Indigenous People in Namibia
 - In English there are 11 words for colors, the Himba have 5. They say that water is “white” and the sky is “black”.





Himba



English

The Himba extend their term for “**black/dark**” (**zoozu**) into the area which is called “purple” in English, including even very bright purple colors.

Source: Knowing color terms enhances recognition: Further evidence from English and Himba, Julie Goldstein, Jules Davidoff, Debi Roberson. In *Journal of Experimental Child Psychology*, 102, 219–238, 2009.

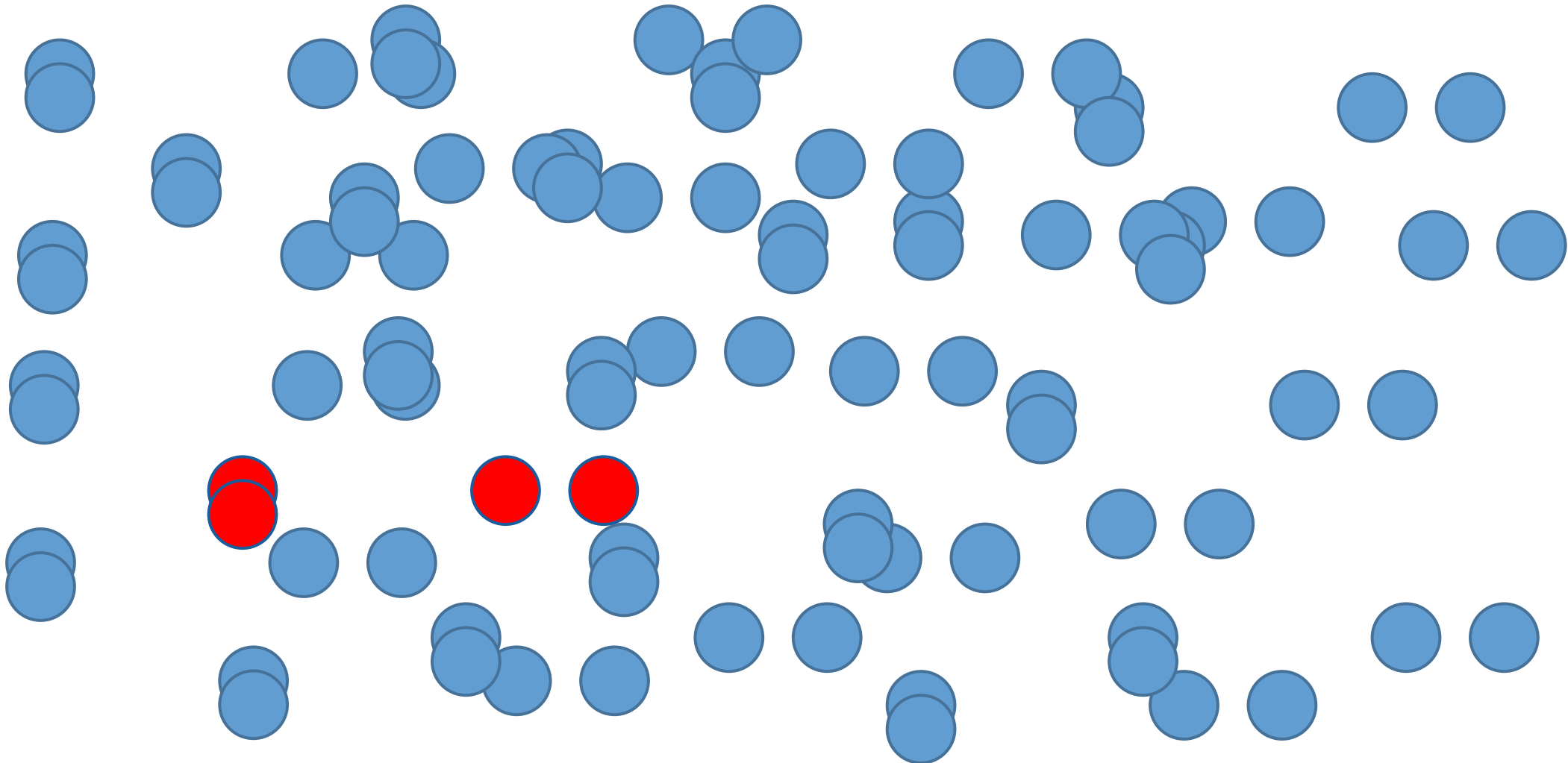
➔ Color perception is influenced by culture.

Chapter
3.6

Pattern Recognition

What do you notice?

44



Where is the man in the beans?

45



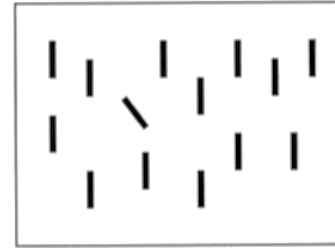
Preattentive Vision / Preattentive Properties

46

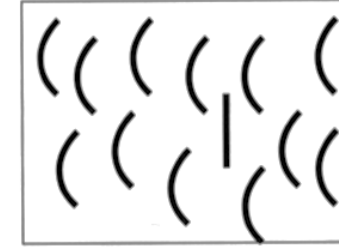
An object with a different preattentive visual property can be quickly detected within a set of similar objects.

- within a fraction of a second (< 200 ms)
- The object needs not to be focused (no fixation).

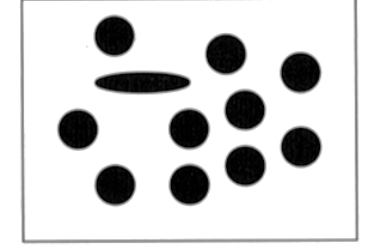
Orientation



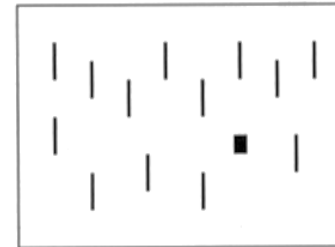
Curved/straight



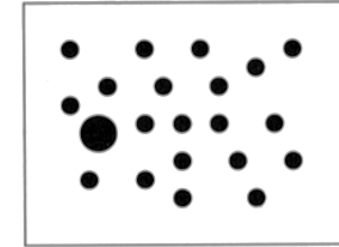
Shape



Shape



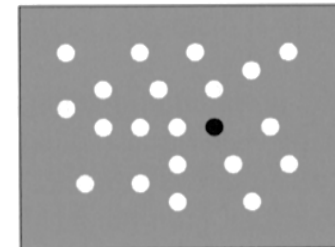
Size



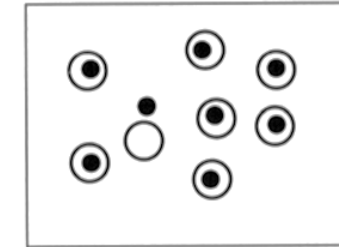
Number



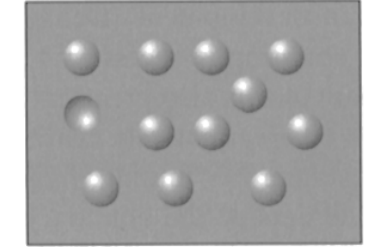
Gray/value



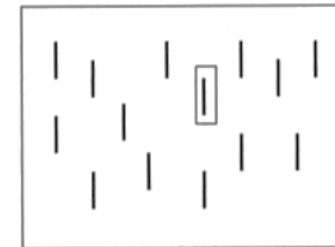
Enclosure



Convexity/concavity



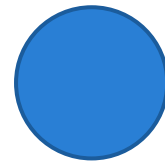
Addition



Motion is preattentive

47

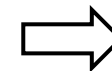
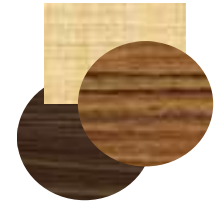
- Motion can be perceived at the border of the periphery (field of view).
- Reflexive eye movement in case of an unexpected motion
- Periodic motions can be masked.



Pattern Recognition

48

- Edge detection
 - ▣ Differences in brightness, color, texture or motion
- Simple patterns
 - ▣ Edges are connected to form contours
 - ▣ Distorted or rotated patterns can be recognized
- Complex patterns: Patterns of patterns
- Object recognition: Compare observed patterns with learned patterns (prior knowledge)

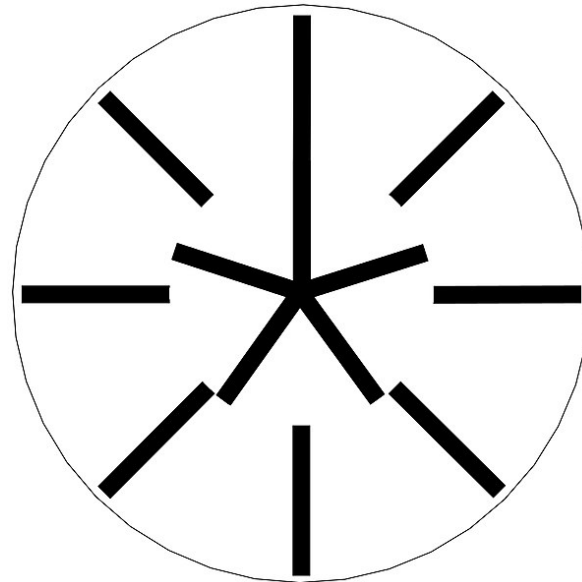


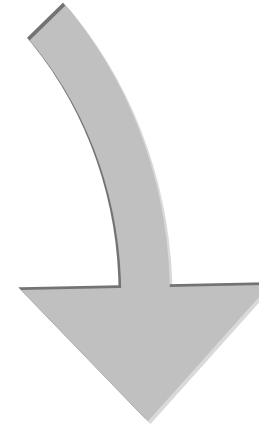
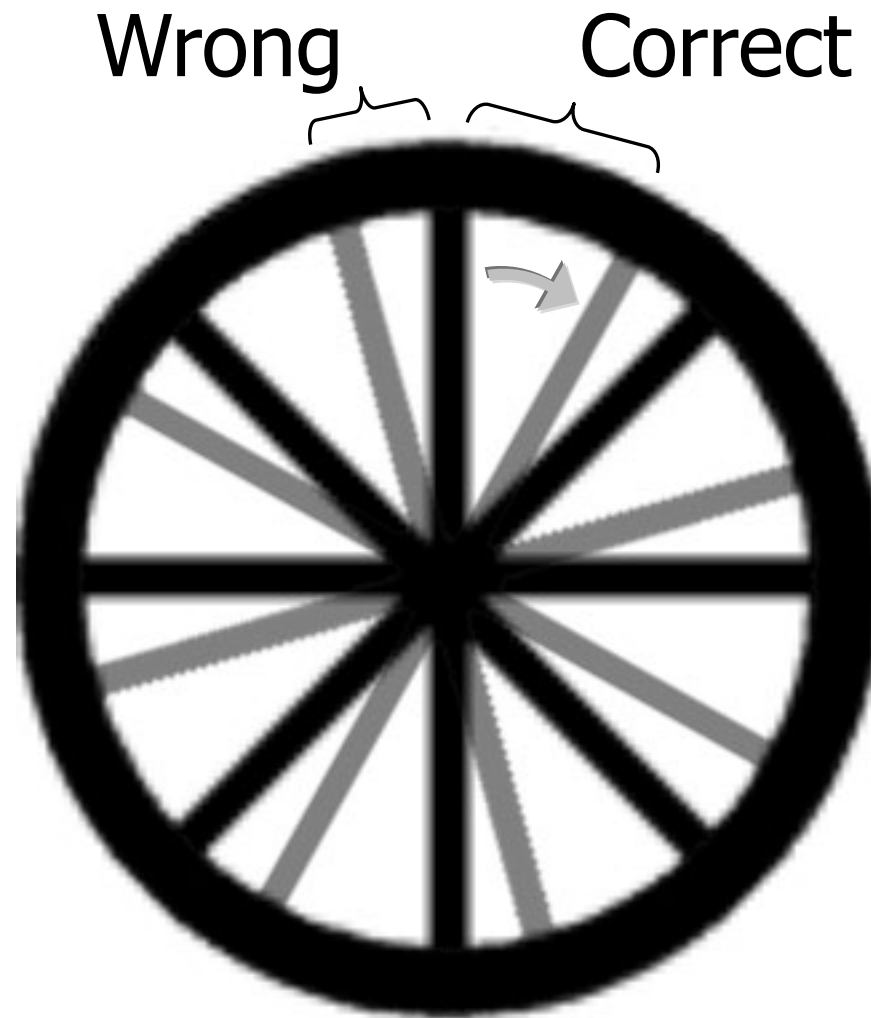
House

Motion recognition



- Question: Do visual elements (e.g., lines, areas) in subsequent images perform the same motion?
 - ▣ Includes pattern recognition in single images (or non moving objects) as discussed above
 - ▣ What does change: form, color, position
 - ▣ Correspondence = What elements in subsequent images correspond to one another (Wagon Wheel Problem)
 - ▣ Same phase (synchronous) is more important than same amplitude (speed)
 - ▣ Causality (delay between the end of one motion and the start of a subsequent motion)



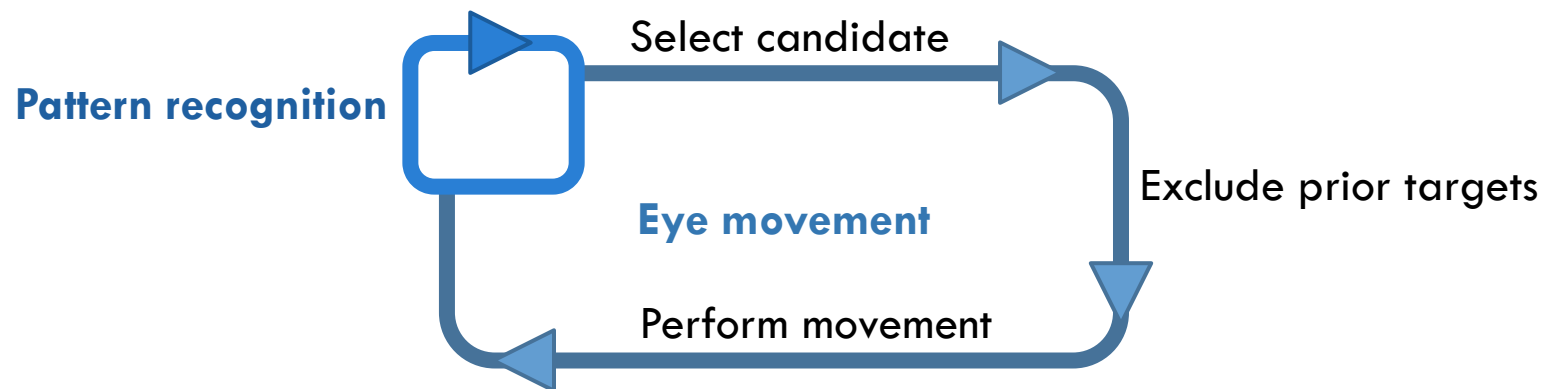


Black: first image
Grey: second image

Visual Search

52

- Eye movement
 - ▣ One to three movements per second
 - ▣ Select a target based on visual properties and prior targets
- Pattern recognition
 - ▣ Recognize pattern at fixation and check whether it is the searched pattern.



Chapter
3.7

Gestalt Laws

Gestalt psychology

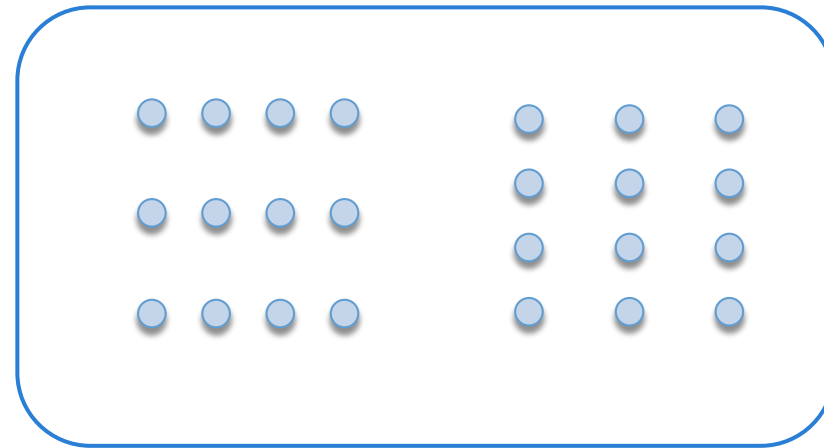
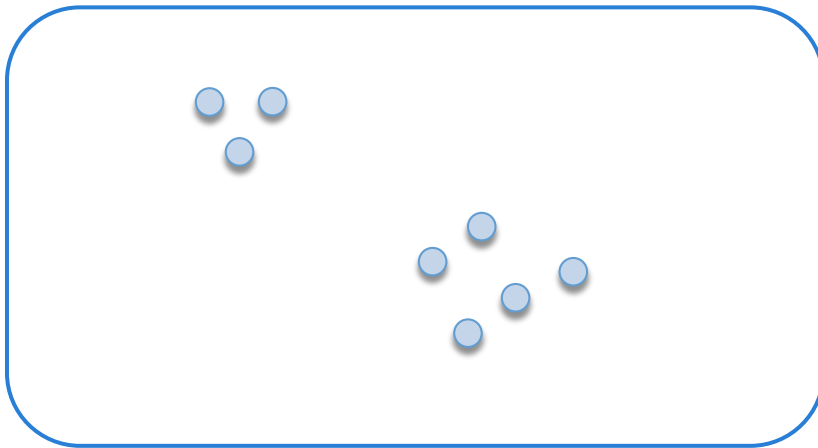
54

- **„Gestalt-Lehre der Psychologie“** (Max Wertheimer, Kurt Koffka, Wolfgang Kohler, et al., 1912): first serious attempts to understand pattern recognition
- **Gestalt Laws:**
 - ▣ Description of properties that let a set of objects appear to belong together.
 - ▣ Additional Gestalt laws were proposed by Stephen Palmer in the 90ties.
 - ▣ **Can be used to represent similarity or relatedness in visualizations.**

Gestalt law: Proximity

55

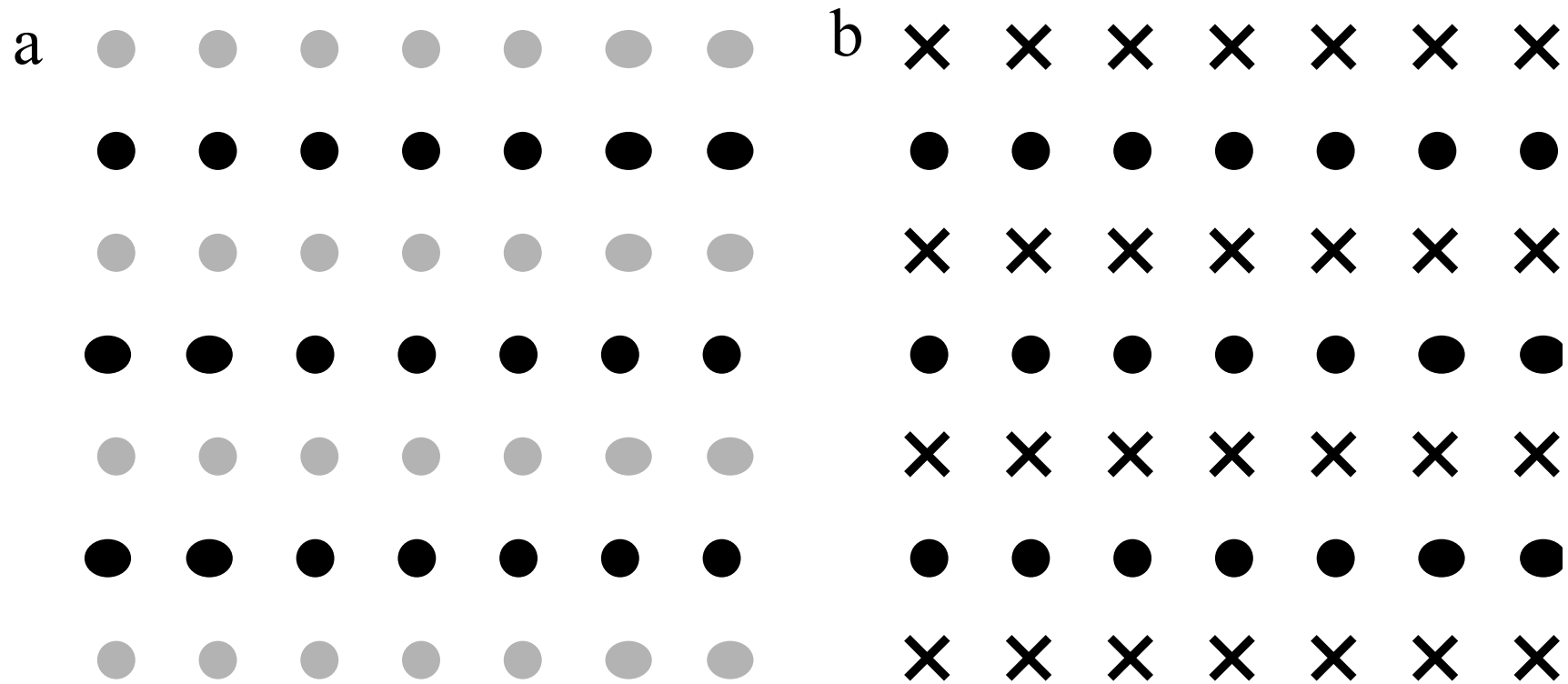
Elements which are placed close to each other are perceived as a group.



Gestalt law: Similarity

56

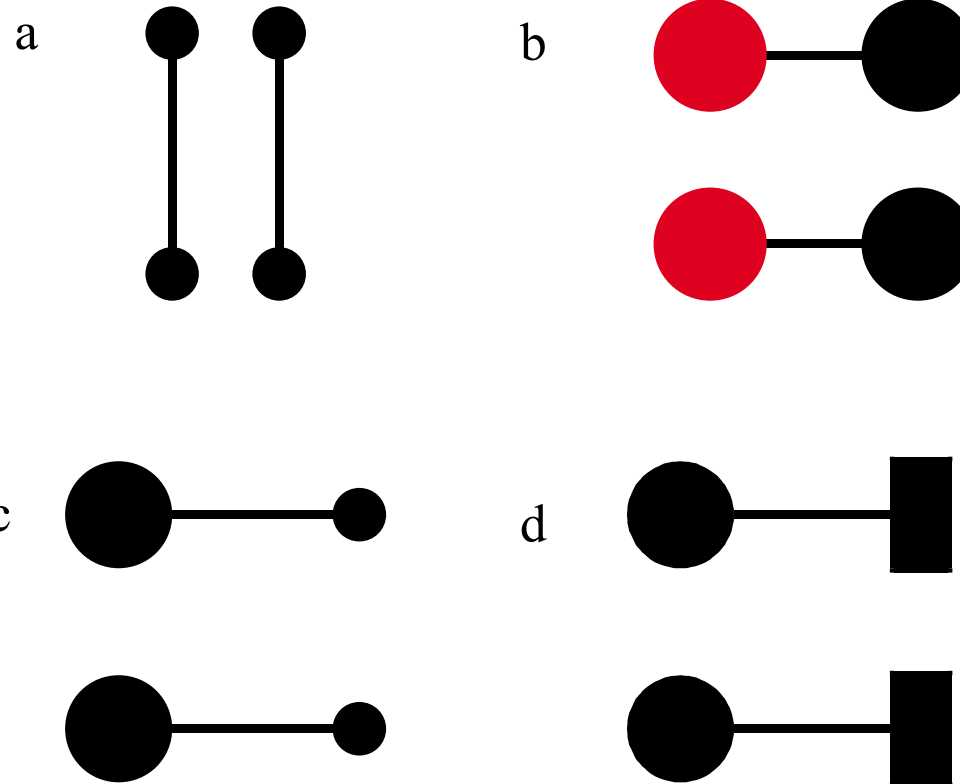
Similar elements (form, color) are perceived as a group.



Gestalt law: Connectedness

57

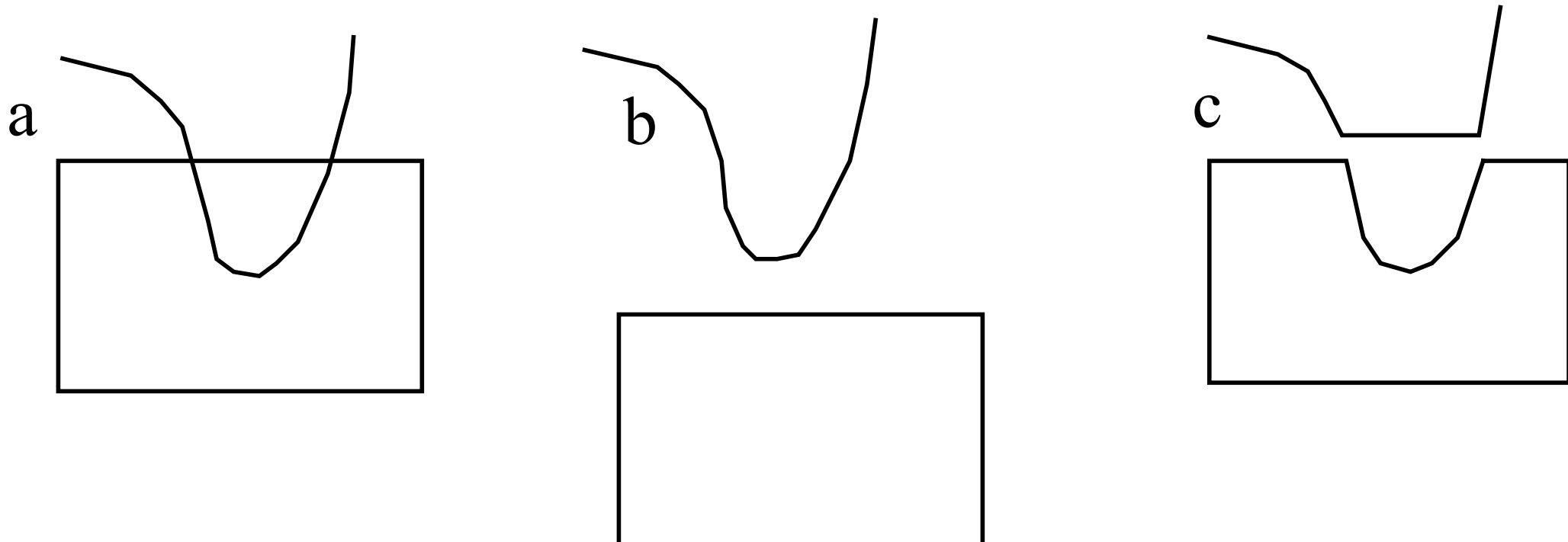
Connected elements are perceived as one object.



Gestalt law: Continuity

58

*For humans it is easier to group **continuous** elements than elements with abrupt changes of direction.*



More Gestalt laws

59

- Law of closure

Lines, that enclose an area, are grouped together. When lines are missing, our perception fills in the visual gap.

- Law of common fate

Two or more elements that move in the same direction are grouped together.

- Law of common region

Elements in a closed region are grouped together.

- Law of simultaneity

Elements that change simultaneously are grouped together.

- ...

Chapter
3.8

Three-Dimensional Perception

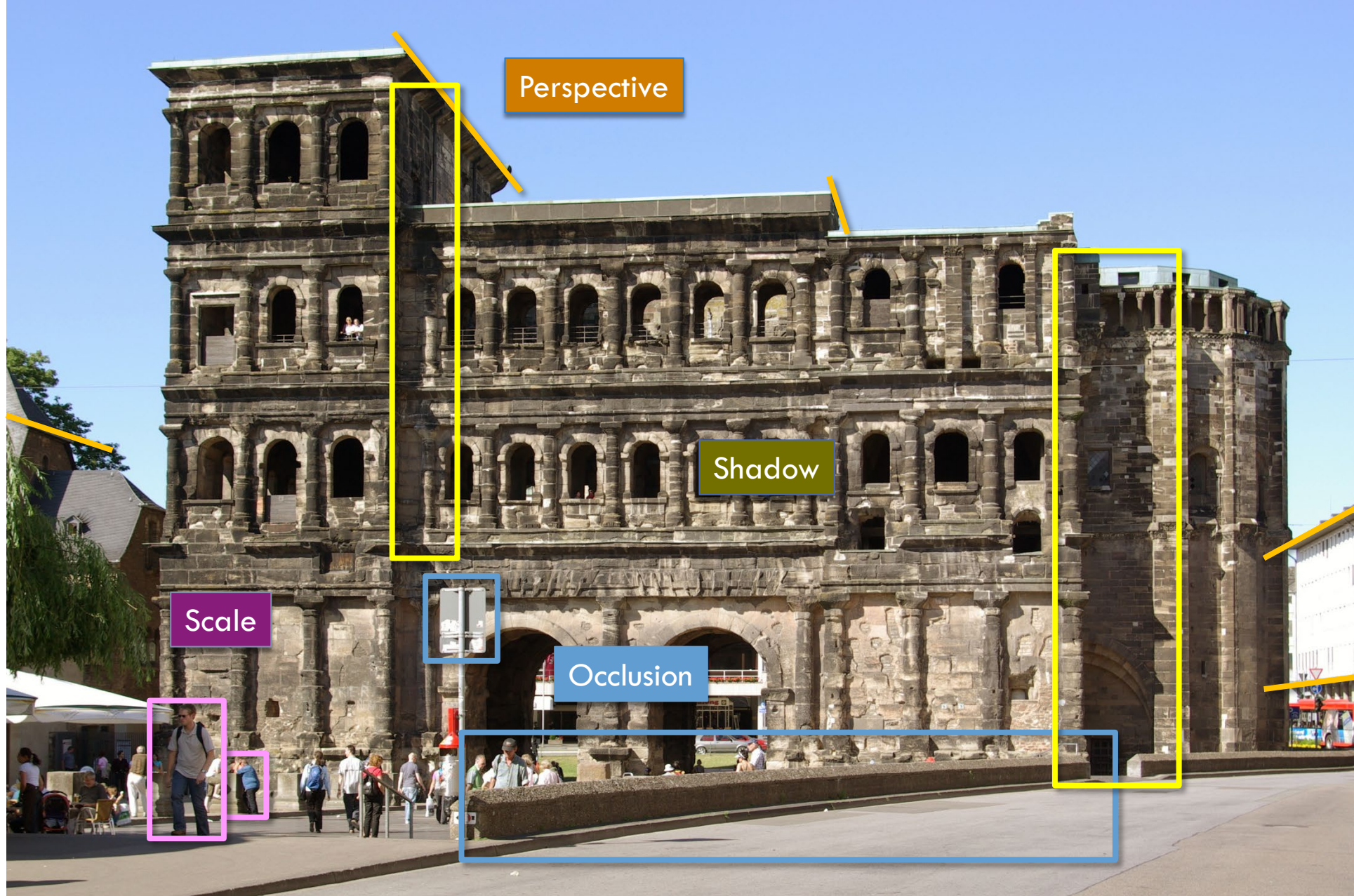


Three-dimensional perception

62

Reconstruction of depth information:

- Stereoscopic vision (in particular at close range)
- Occlusion of objects
- More depth cues: depth of field (Tiefenschärfe), perspective, shadow, scale, contrast, motion parallax (how near and far objects will move across the retina of an eye as we move along in the world)
- Prior knowledge

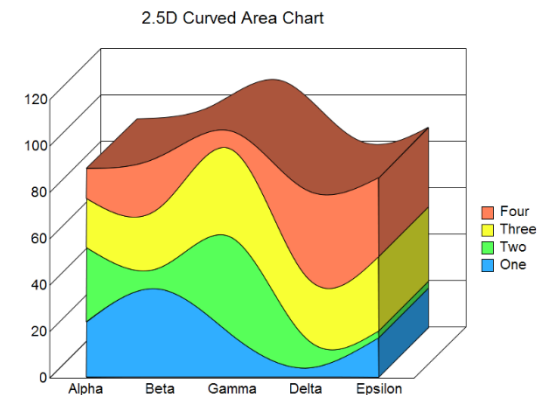


3D Information Visualization

64

- 3D questionable for visualizing abstract data
- Major problems: occlusion, navigation in 3D
- 2.5D visualizations:
 - ▣ tries to combine advantages of 2D and 3D
 - ▣ indicates 3rd dimension only by shadow, perspective, etc.
 - ▣ avoids occlusion of objects, e.g. by using transparency

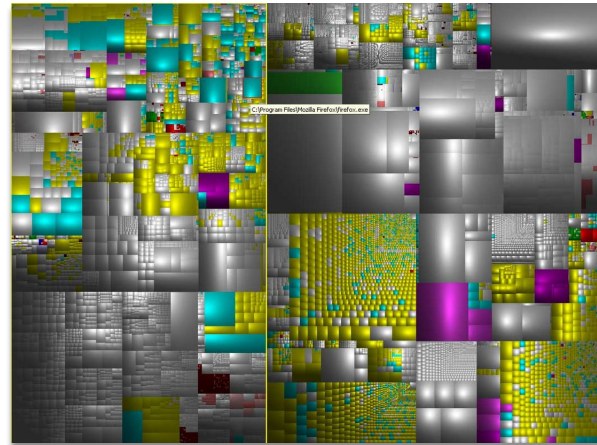
Parallax
scrolling



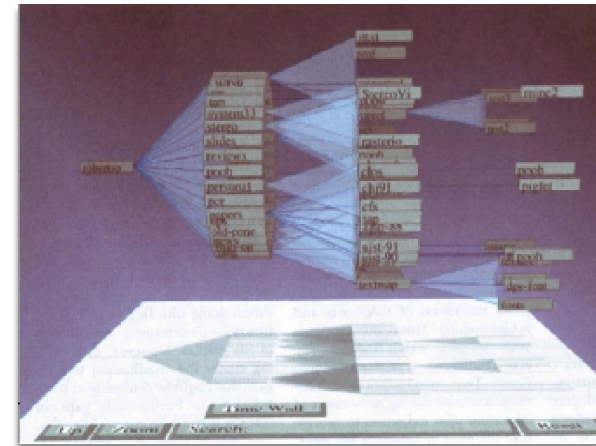
Examples of 2.5D visualizations

65

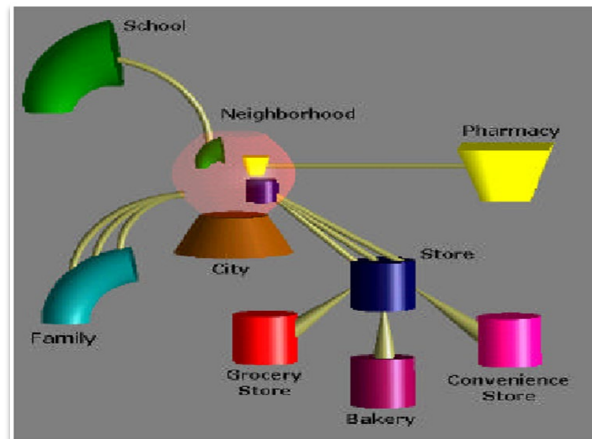
Cushion
Treemap



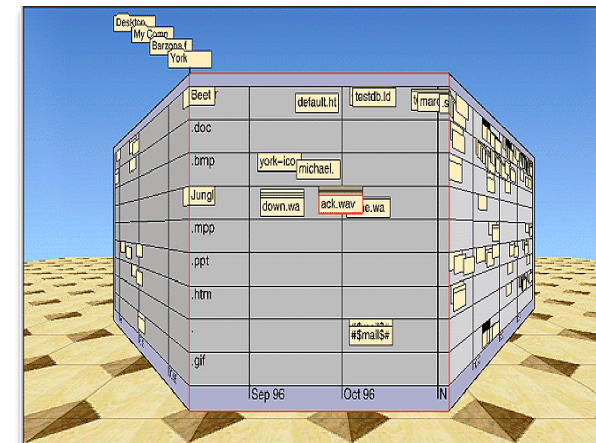
Cone
Trees



UML
Geons



Perspective
Wall



Conclusion

Design recommendations



67

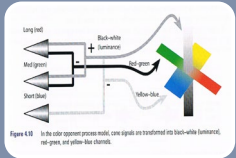
- Color perception in the center, not in the periphery
 - **Color coding for focused objects**
- Motion in the periphery triggers eye movement (orientation towards the motion)
 - **Animated “Eyecatcher” on the edge of the screen**
 - **Common motion can be used to indicate relatedness of distant objects**
- Color blindness
 - **Use color palettes with yellow/blue variations**
 - see f.e. <http://colorbrewer2.org>
- Inertia of the eye
 - **Avoid strong colors and contrasts. Long focussing can lead to ghost images.**

Summary

68



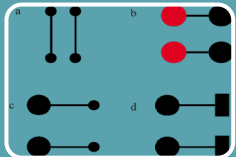
Visual perception does not work like a camera, but is selective and interpreting.



Perceived colors are composed and relative to the environment.



Visual perception includes pattern recognition.



Gestalt laws describe what visual properties help to group visual elements together.



Three-dimensional perception is rather complex and renders 3D visualizations of abstract data questionable.